

Hypothesis closure and pre-registered predictions vs observed outcomes

Hypothesis / prediction

Observed closure

Evidence type



H1 strong

FALSIFIED

frozen recipe direct transfer



H2 medium

FALSIFIED BY STRUCTURAL DEFINITION

partial recalibration



H3 surrogate

SUPPORTED ON N3

full Stage A/B/C RMSE



H3 controller

FALSIFIED WITH REGIME DEPENDENT FAILURE MODES

controller full regime



Stretch none

expected FAIL → observed PASS

commercial mode=none



Stretch C_zon

expected scale-dependent → observed 1.909×

commercial C_zon



Stretch RMSE

expected 50–90% → observed 87.81%

commercial full recalibration